

INDIAN STATISTICAL INSTITUTE, BANGALORE CENTRE

B.MATH - Second Year, First Semester, 2025-26

Introduction to Statistical Inference, Final Examination,

12th November, 2025

Answer All Questions (Q1 is Compulsory)

Maximum Marks: 50

Time: 3 hours

1. Define the following:

- (a) Sufficient Statistic
- (b) Exponential Families
- (c) UMVUE
- (d) Monotone Likelihood Ratio
- (e) Credible Set

[5 × 2]

2. For observations $Y_1, \dots, Y_n$, consider the linear model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where x_i is the value of a co-variate corresponding to Y_i and ϵ_i are i.i.d. errors having the $N(0, \sigma^2)$ distribution. Here β_0, β_1 and $\sigma^2 > 0$ are unknown parameters and x_i are treated as known constants.

- (a) Show that the distribution of $Y_1, \dots, Y_n$ belongs to k -parameter exponential family. Find k .
- (b) Find the MLEs of $(\beta_0, \beta_1, \sigma^2)$?

[4+6]

3. Let $X \sim N_p(\mu, \Sigma)$, $\Sigma > 0$ (i.e., positive definite), and let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix},$$

where X_1 and μ_1 are of length k . Also, let

$$\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

Then show that

- (i) $\Sigma_{11.2} > 0$ and $X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \sim N_k(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11.2})$ and is independent of X_2 ;
- (ii) The conditional distribution of X_1 given X_2 is

$$N_k(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(X_2 - \mu_2), \Sigma_{11.2}).$$

[5+5]

4. Let $X_1, \dots, X_n$ be i.i.d. from $f(x; \theta) = \theta e^{-\theta x}$, $x > 0$ (rate parameter $\theta > 0$).

- (a) Show that the MLE $\hat{\theta}_{\text{MLE}} = \frac{n}{\sum X_i}$ is consistent and asymptotically normal.
- (b) Show that the estimator of the mean $1/\theta$, given by $\bar{X}$, is unbiased and attains the Cramer-Rao lower bound for $1/\theta$. Is this estimator efficient?

[3+3+1+2+1]

5. Let $X_1, X_2, \dots, X_n$ be i.i.d. from the probability density function

$$f(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0,$$

where θ is a scale parameter.

- (a) Does the family have MLR in any simple statistic T for testing larger versus smaller values of θ ?
- (b) Using the MLR property, find a *Uniformly Most Powerful (UMP)* level- α test for

$$H_0 : \theta \leq \theta_0 \quad \text{versus} \quad H_1 : \theta > \theta_0.$$

- (c) Express the rejection region in terms of T .
- (d) Does there exist UMP test for testing

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta \neq \theta_0.$$

Justify your answer.

[3+3+2+2]

6. Let $X_1, X_2, \dots, X_m$ and $Y_1, Y_2, \dots, Y_n$ be independent random samples, respectively, from $\text{Poisson}(\lambda)$ and $\text{Poisson}(3\lambda)$, where $\lambda > 0$ is unknown.

- (a) Does this model belong to the exponential family of distributions? Justify.
- (b) Find the minimal sufficient statistic $T = T(X, Y)$ for λ .
- (c) Find the UMVUE of λ .

[3+3+4]